

SECTION 7 - TABLES, CHARTS AND PROFILES

INDEX

Sub-Section 1

Sag Charts
Copper Conductors

Subject

Conductor	Basic Span	Regulations	Max. Tension	Drawing No.	
0.20 (7/.193)	500 ft.	El.C.53	5630 lbf	Y003X 7101	*
0.20 (7/.193)	500 ft.	El.C.53	4265 lbf	Y003X 7102	*
0.20 (7/.193)	300 ft.	El.C.53	5650 lbf	Y003X 7103	*
0.15 (7/.166)	700 ft.	El.C.53	4263 lbf	Y003X 7104	*
0.15 (7/.166)	500 ft.	El.C.53	4263 lbf	Y003X 7105	*
0.15 (7/.166)	400 ft.	El.C.53	4263 lbf	Y003X 7106	*
0.15 (7/.166)	360 ft.	El.C.53	4263 lbf	Y003X 7107	*
0.10 (7/.136)	700 ft.	El.C.53	2936 lbf	Y003X 7108	*
0.10 (7/.136)	500 ft.	El.C.53	2936 lbf	Y003X 7109	*
0.10 (7/.136)	400 ft.	El.C.53	2936 lbf	Y003X 7110	*
0.05 (3/.147)	350 ft.	El.C.53	1460 lbf	Y003X 7111	*
0.05 (3/.147)	300 ft.	El.C.53	1460 lbf	Y003X 7112	*
0.05 (3/.147)	300 ft.	BS.1320	1170 lbf	Y003X 7113	*
0.025 (3/.104)	250 ft.	BS.1320	608 lbf	Y003X 7114	*
0.08 (7/.122)	500 ft.	El.C.53	2348 lbf	Y003X 7115	*
0.05 (3 SWG)	350 ft.	BS.1320	1150 lbf	Y003X 7116	*
0.05 (3 SWG)	300 ft.	El.C.53	1437 lbf	Y003X 7117	*
0.035 (5 SWG)	250 ft.	BS.1320	823 lbf	Y003X 7118	*
0.029 (6 SWG)	250 ft.	BS.1320	693 lbf	Y003X 7119	*
0.025 (7 SWG)	250 ft.	BS.1320	591 lbf	Y003X 7120	*
0.10 (7/.136)	} LV Sag curves based on 3.1 ft. sag at 122° F in 150 ft. span.			Y003X 7121	*
0.05 (3 SWG)					
0.029 (6 SWG)					

Added
Sept. '71

* These sag charts only to be used for checking sag of existing conductors.
For erection of new conductors or re-erection of existing conductors refer
to Sag Tables, listed in Section 7.5.

Sub-Section 2

Sag Charts, Aluminium and
Aluminium Alloy Conductors

<u>Subject</u>				<u>Drawing No.</u>	
Conductor	Basic Span	Regulations	Max. tension		
0.15 (19/.137) Al. Alloy	800 ft.	El.C.53	4500 lbf.	Y003X 7201	*
0.15 (19/.137) Al. Alloy	500ft.	"	4500 "	Y003X 7202	*
0.15 (19/.137) Al. Alloy	400 "	"	4200 "	Y003X 7203	*
0.10 (7/.183) Al. Alloy	500 "	"	3455 "	Y003X 7204	*
0.10 (7/.183) Al. Alloy	400 "	"	3040 "	Y003X 7205	*
0.10 (7/.183) Al. Alloy	400 "	"	3455 "	Y003X 7206	*
0.05 (7/.130) Al. Alloy	350 "	"	1920 "	Y003X 7207	*
0.05 (7/.130) Al. Alloy	350 "	"	1746 "	Y003X 7208	*
0.05 (7/.130) Al. Alloy	350 "	BS1320	1536 "	Y003X 7209	*
0.04 (7/.116) Al. Alloy	350 "	"	1220 "	Y003X 7210	*
0.04 (3/.178) Al. Alloy	350 "	"	1244 "	Y003X 7211	*
0.025 (7/.092) Al. Alloy	300 "	"	765 "	Y003X 7212	*
0.025 (3/.141) Al. Alloy	300 "	"	780 "	Y003X 7213	*
0.015 (3/.111) Al. Alloy	250 "	"	484 "	Y003X 7214	*
0.10 (19/.110) Al. Alloy	400 "	El.C.53	1900 "	Y003X 7215	*
0.10 (19/.110) Al. Alloy	400 "	El.C.53	3477	Y003X 7216	*
0.1 (7/.173 Aluminium (LV conductor)				Y003X 7217	*

Added
Sept.'71

* These sag charts only to be used for checking sag of existing conductors. For erection of new conductors or re-erection of existing conductors refer to Sag Tables, listed in Section 7.5.

Sub-section 3

Sag Chart. ACSR
Conductors

Subject

Drawing No.

Conductor :	Basic Span	Max. Working Tension		
0.175 sq. ins CE (NEW)	700 ft.	6200 lbf	Y003X 7301	*
0.175 sq. ins. CE (OLD)	700 ft.	5843 "	Y003X 7302	*
0.075 sq. ins.-- CE (NEW)	500 "	6095 "	Y003X 7303	*
0.175 sq. ins. CE (OLD)	500 "	5843 "	Y003X 7304	*
0.175 sq. ins. CE (NEW)	400 "	5622 "	Y003X 7305	*
0.175 sq. ins. CE (OLD)	400 "	5394 "	Y003X 7306	*
0.175 sq. ins. CE (NEW)	300 "	5216 "	Y003X 7307	*
0.175 sq. ins. CE (OLD)	300 "	5008 "	Y003X 7308	*
0.15 sq. ins. CE (NEW)	500 "	5269 "	Y003X 7309	*
0.15 sq. ins. CE (OLD)	500 "	5057 "	Y003X 7310	*
0.15 sq. ins. CE (NEW)	300 "	4505 "	Y003X 7311	*
0.15 Sq. in. CE (OLD)	300 "	4334 "	Y003X 7312	*
0.10 sq. ins. CE (NEW)	600 "	3770 "	Y003X 7313	*
0.10 sq. ins. CE (OLD)	600 "	3650 "	Y003X 7314	*
0.025 sq. ins. CE (OLD)	350 "	863 "	Y003X 7315	*

Sub-section 4

Sag Charts. Miscellaneous

Subject

Drawing No.

Conductor	Basic Span	Max. Working Tension		
0.10 steel cored copper	700 ft	4406 lbf	Y003X 7401	*
0.10 " "	300 "	4406 "	Y003X 7402	*
7/11½ steel	350 "	2962 "	Y003X 7403	*
0.05(3/.158)	400 "	2333 "	Y003X 7404	*
Cd. Cu.				
9 pair 1/.044 pilot	250 "	3700 "	Y003X 7405	*
0.0225 split concentric	90 "	250 "	Y003X 7406	*
0.04 split concentric	90 ft	250 lbf	Y003X 7407	*
0.15 sq. ins. steel cored copper	650 ft	12000 lbf	Y003X 7408	*

7.5 INDEX OF SAG TABLES

IMPERIAL

Tables based on Electricity (Overhead Lines) Regulations No. 1355 (1970).

Specification	Conductor	Basic Span ft.	Factor of Safety	Sag Table Ref.
BEBS-L1	32mm ² Copper	150	2.92	1
	70mm ² Copper	150	4.35	2
	0.05in ² Copper	150	2.92	11 *
	0.10in ² Copper	150	4.35	12 *
BS.1320	50mm ² Al. Alloy	350	2.5	3
	0.05in ² Al. Alloy	350	2.5	13 *
	16mm ² Copper	250	2.5	14 *
	0.025in ² Copper	250	2.5	15 *
	32mm ² Copper	300	2.5	16 *
	0.05in ² Copper	300	2.5	17 *
CE/C/31	100mm ² Al. Alloy	300	2.5	4
	0.10in ² Al. Alloy	300	2.5	18 *
	0.05in ² Al. Alloy	300	2.5	19 *
	32mm ² Copper	350	2	20 *
	0.05in ² Copper	350	2	21 *
	70mm ² Copper	350	2	22 *
	0.10in ² Copper	350	2	23 *
	50mm ² Al. Alloy	300	2.5	43 *
	150mm ² Al. Alloy	300	2.5	44 *
	100mm ² Copper	350	2	45 *
CE/C/36	New 175mm ² ACSR	250	3.5 but + 8%	5
	Old 175mm ² ACSR	250	3.5	6
	New 175mm ² ACSR	300	3.5 but + 8%	58 *
	Old 175mm ² ACSR	300	3.5	59 *
	New 175mm ² ACSR	350	3.5 but + 8%	60 *
	Old 175mm ² ACSR	350	3.5	61 *
	Old 0.175in ² ACSR	250	3.5	24 *
	Old 0.15in ² ACSR	250	3.5	25 *
	70mm ² Copper	300	2	26 *
	0.10in ² Copper	300	2	27 *
	100mm ² Copper	300	2	28 *
	0.15in ² Copper	300	2	29 *
	New 150mm ² ACSR	250	3.5 but + 8%	46 *
	Old 150mm ² ACSR	250	3.5	47 *

Added
May 74

Specification	Conductor	Basic Span Ft.	Factor of Safety	Sag Table Ref.
CE/C/37	New 175 mm ² ACSR	500	3 but + 8%	7
	Old 175 mm ² ACSR	500	3	8
	Old 0.175 in ² ACSR	500	3	30 *
	Old 0.15 in ² ACSR	500	3	31 *
	70 mm ² Copper	500	2	32 *
	0.10 in ² Copper	500	2	33 *
	100 mm ² Copper	500	2	34 *
	0.15 in ² Copper	500	2	35 *
	New 150 mm ² ACSR	500	3 but + 8%	48 *
	Old 150 mm ² ACSR	500	3	49 *
CE/OM/DCI	New 175 mm ² ACSR	400	3.25 but + 8%	9
	Old 175 mm ² ACSR	400	3.25	10
	Old 0.175 in ² ACSR	400	3.25	36 *
	100 mm ² Copper	400	2	50 *
Services	16 mm ² Split Conc.	90	-	37
	25 mm ² Split Conc.	90	-	38
	16 mm ² Concentric	90	-	39 *
	25 mm ² Concentric	90	-	40 *
	0.0225 in ² Split Conc.	90	-	41 *
	0.04 in ² Split Conc.	90	-	42 *
	25 mm ² Conc. (Al. phase)	90	-	56
	35 mm ² Conc. (Al. phase)	90	-	57
Miscellaneous	New 175 mm ² ACSR	700	3 but + 8%	51 *
	Old 175 mm ² ACSR	700	3	52 *
	0.15 in ² Copper	600	2	53 *
	New 175 mm ² ACSR	600	3 but + 8%	54 *
	Old 175 mm ² ACSR	600	3	55 *
	Old 0.175 in ² ACSR	700	2.5	58 *
	Old 0.175 in ² ACSR	1000	2.5	59 *

* Sag tables not included in the Overhead Line Manual-Part 1 but, if required, copies may be obtained on request to the Chief Engineer.

A computer program has been written to evaluate Sags and Tensions. If Sag Tables additional to those listed above are required, they may also be obtained on request to the Chief Engineer.

7.6 INDEX OF ORDINATE TABLES

IMPERIAL

Tables based on Electricity (Overhead Lines) Regulations No. 1355 (1970)

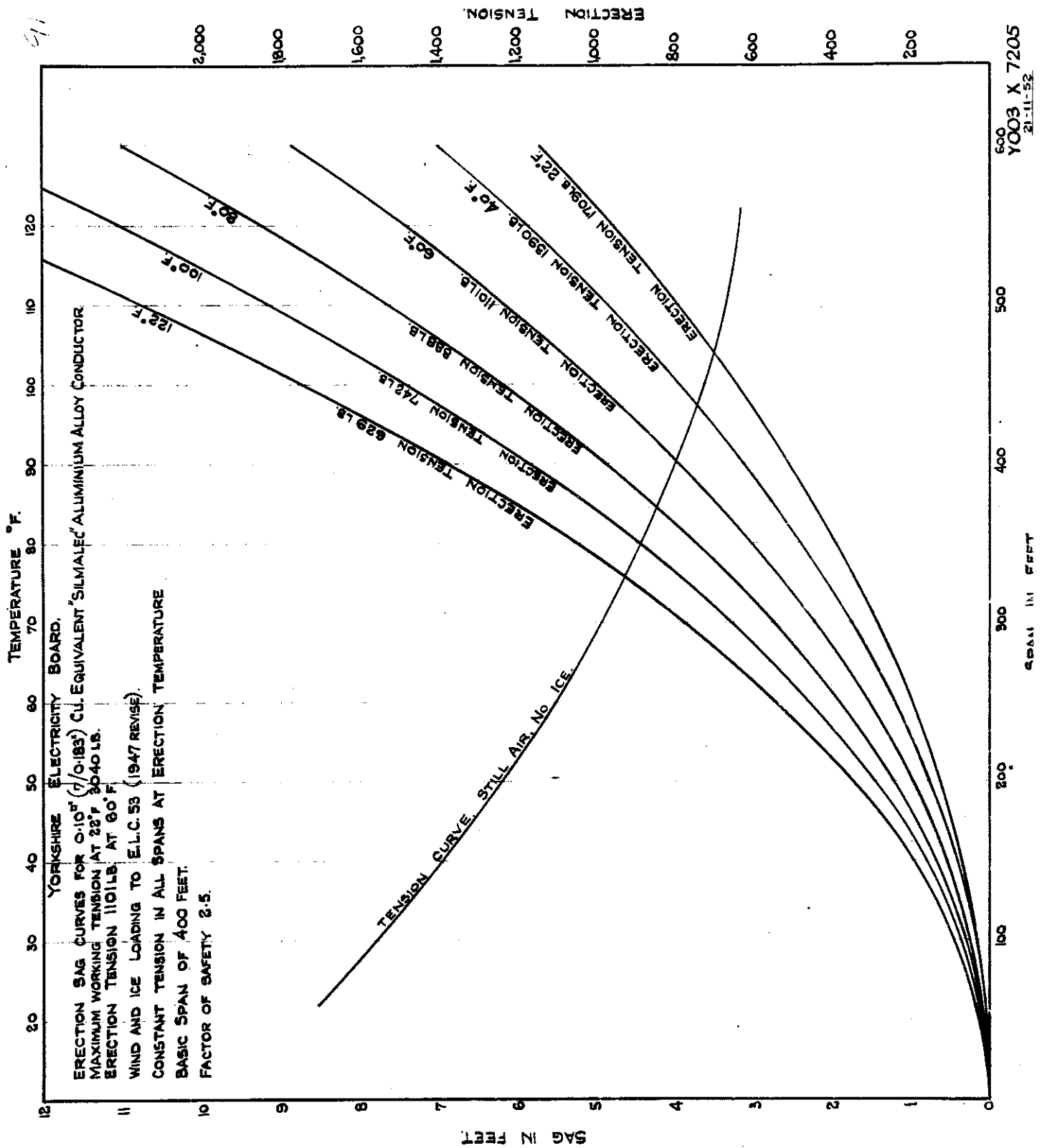
Specification	Conductor	Basic Span ft.	Factor of Safety	Ordinate Table Ref.
BEBS-L1	32mm ² Copper	150	2.92	1
	70mm ² Copper	150	4.35	2
	0.05in ² Copper	150	2.92	8
	0.10in ² Copper	150	4.35	9
BS.1320	50mm ² Al. Alloy	350	2.5	3
	0.05in ² Al. Alloy	350	2.5	10
	16mm ² Copper	250	2.5	11
	0.025in ² Copper	250	2.5	12
	32mm ² Copper	300	2.5	13
	0.05in ² Copper	300	2.5	14
CE/C/31	100mm ² Al. Alloy	300	2.5	4
	0.10in ² Al. Alloy	300	2.5	15
	0.05in ² Al. Alloy	300	2.5	16
	32mm ² Copper	350	2	17
	0.05in ² Copper	350	2	18
	70mm ² Copper	350	2	19
	0.10in ² Copper	350	2	20
	50mm ² Al. Alloy	300	2.5	34
	150mm ² Al. Alloy	300	2.5	35
	100mm ² Copper	350	2	36
CE/C/36	175mm ² ACSR	250	3.5	5
	175mm ² ACSR	300	3.5	43
	175mm ² ACSR	350	3.5	44
	0.175in ² ACSR	250	3.5	21
	0.15in ² ACSR	250	3.5	22
	70mm ² Copper	300	2	23
	0.10in ² Copper	300	2	24
	100mm ² Copper	300	2	25
	0.15in ² Copper	300	2	26
	150mm ² ACSR	250	3.5	37

Added
May 74

Specification	Conductor	Basic Span Ft.	Factor of Safety	Ordinate Table Ref.
CE/C/37	175 mm ² ACSR	500	3	6
	0.175 mm ² ACSR	500	3	27
	0.15 in ² ACSR	500	3	28
	70 mm ² Copper	500	2	29
	0.10 in ² Copper	500	2	30
	100 mm ² Copper	500	2	31
	0.15 in ² Copper	500	2	32
	150 mm ² ACSR	500	3	38
CE/OM/DC1	175 mm ² ACSR	400	3.25	7
	0.175 in ² ACSR	400	3.25	33
	100 mm ² Copper	400	2	39
Miscellaneous	175 mm ² ACSR	700	3	40
	0.15 in ² Copper	600	2	41
	175 mm ² ACSR	600	3	42
	0.175 in ² ACSR	700	2.5	43
	0.175 in ² ACSR	1000	2.5	44

Added

Copies of the Ordinate Tables are not included in the Overhead Line Manual but, if required, they may be obtained on request to the Chief Engineer.



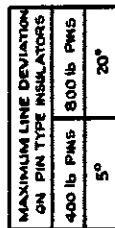
YORKSHIRE ELECTRICITY BOARD
SAG CURVES 0.05" COPPER EQUIVALENT (7/130") SILMALEC ALUMINIUM

ALLOY CONDUCTOR

NO ICE TO B.S. 1320/1946

CONSTANT TENSION IN ALL SPANS AT ERECTION TEMPERATURE

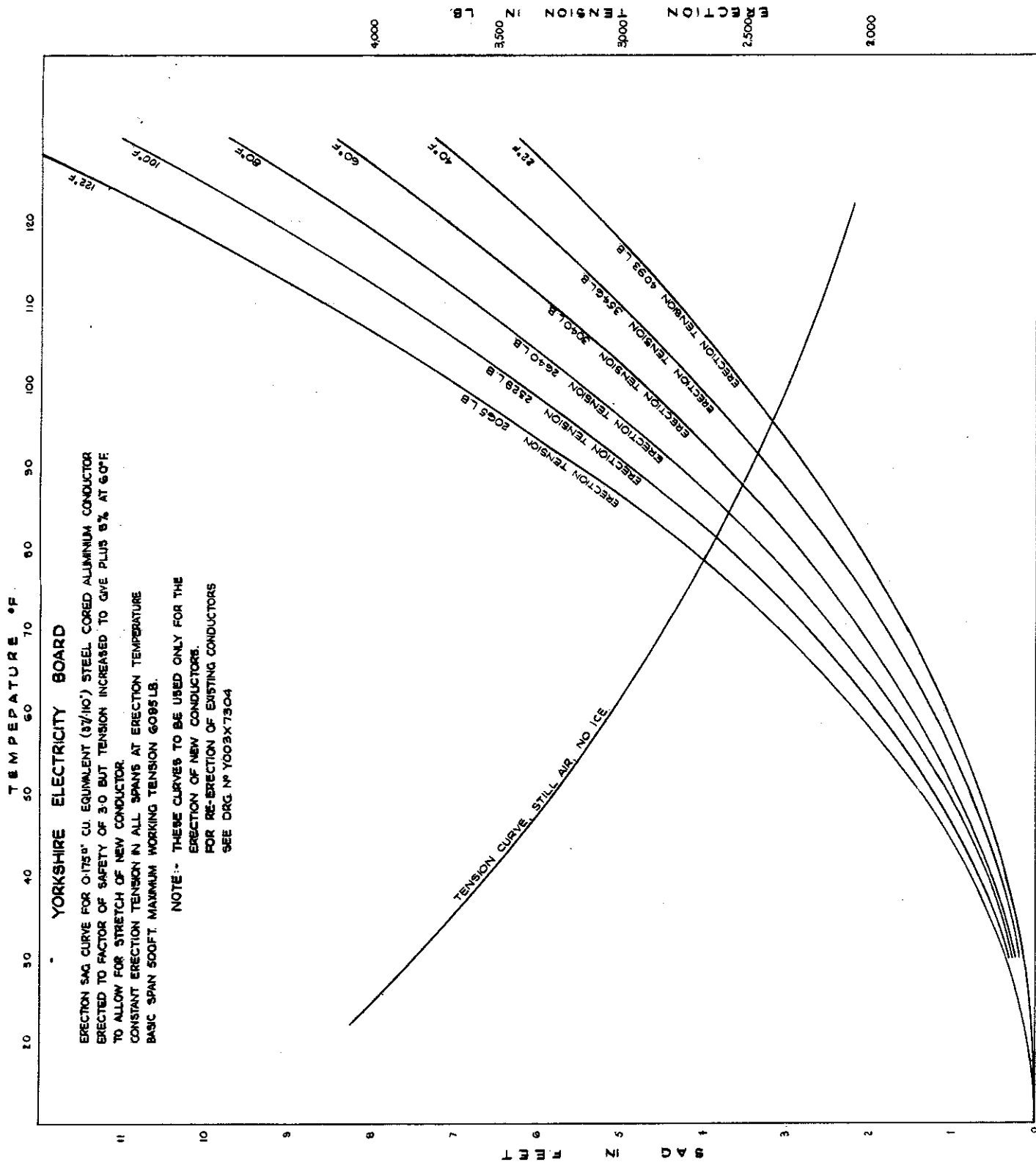
BASIC SPAN 350 FT.

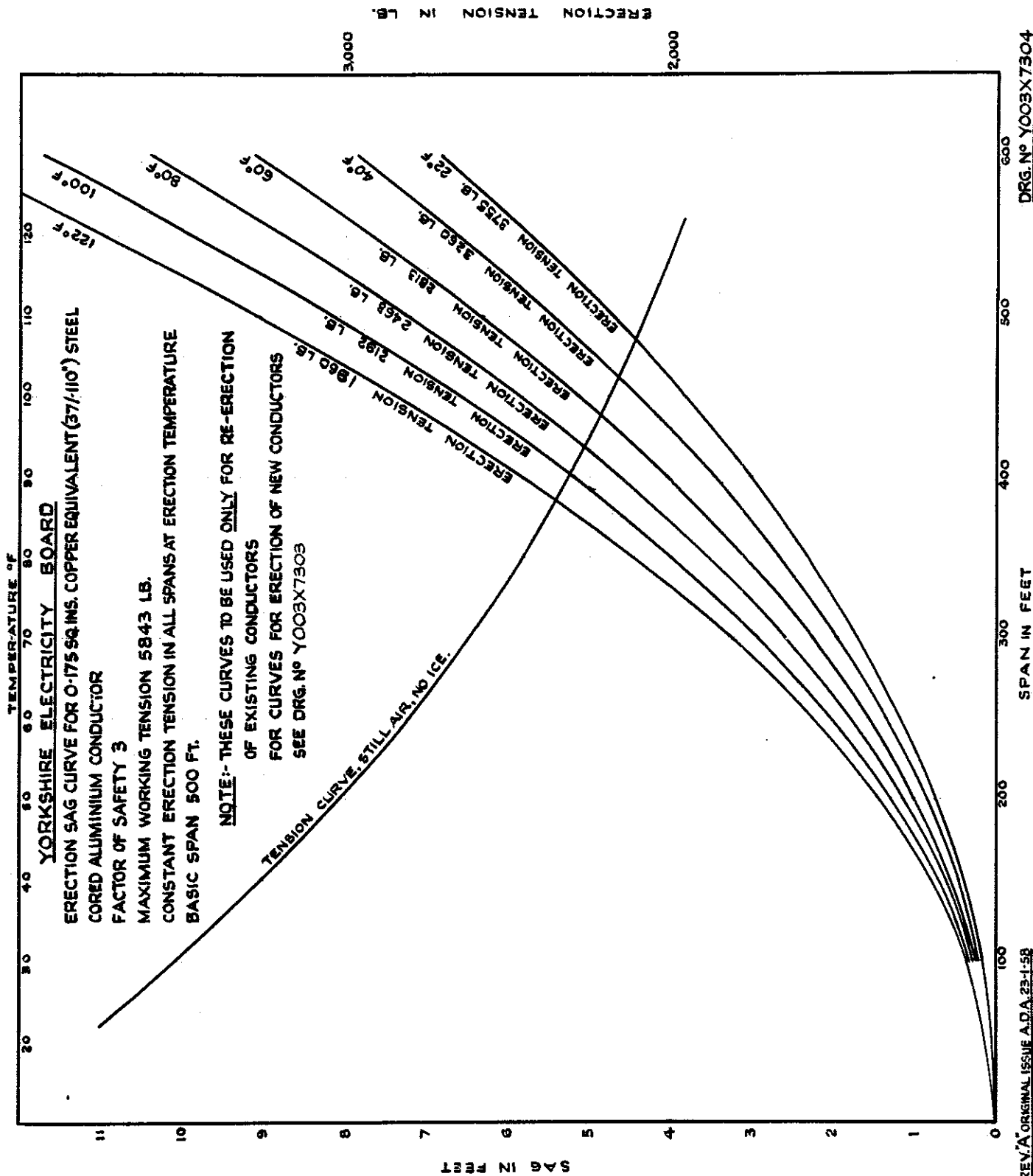


YORKSHIRE ELECTRICITY BOARD

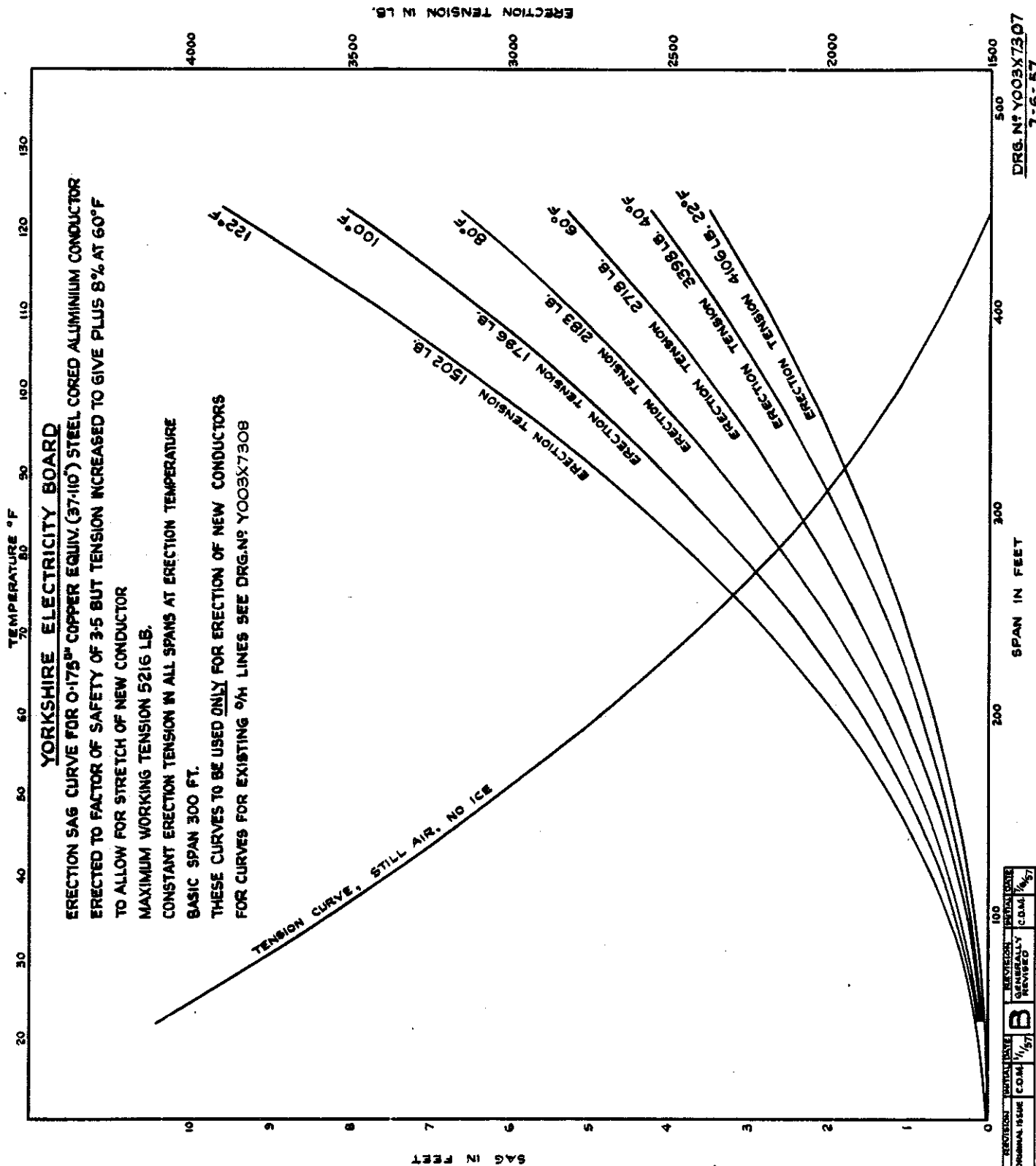
ERECTION SAG CURVE FOR 0.175" CU. EQUIVALENT (87/10") STEEL CORED ALUMINUM CONDUCTOR
 ELECTED TO FACTOR OF SAFETY OF 3.0 BUT TENSION INCREASED TO GIVE PLUS 5% AT 60°F.
 TO ALLOW FOR STRETCH OF NEW CONDUCTOR.
 CONSTANT ERECTION TENSION IN ALL SPANS AT ERECTION TEMPERATURE
 BASIC SPAN 500FT. MAXIMUM WORKING TENSION 6095 LB.

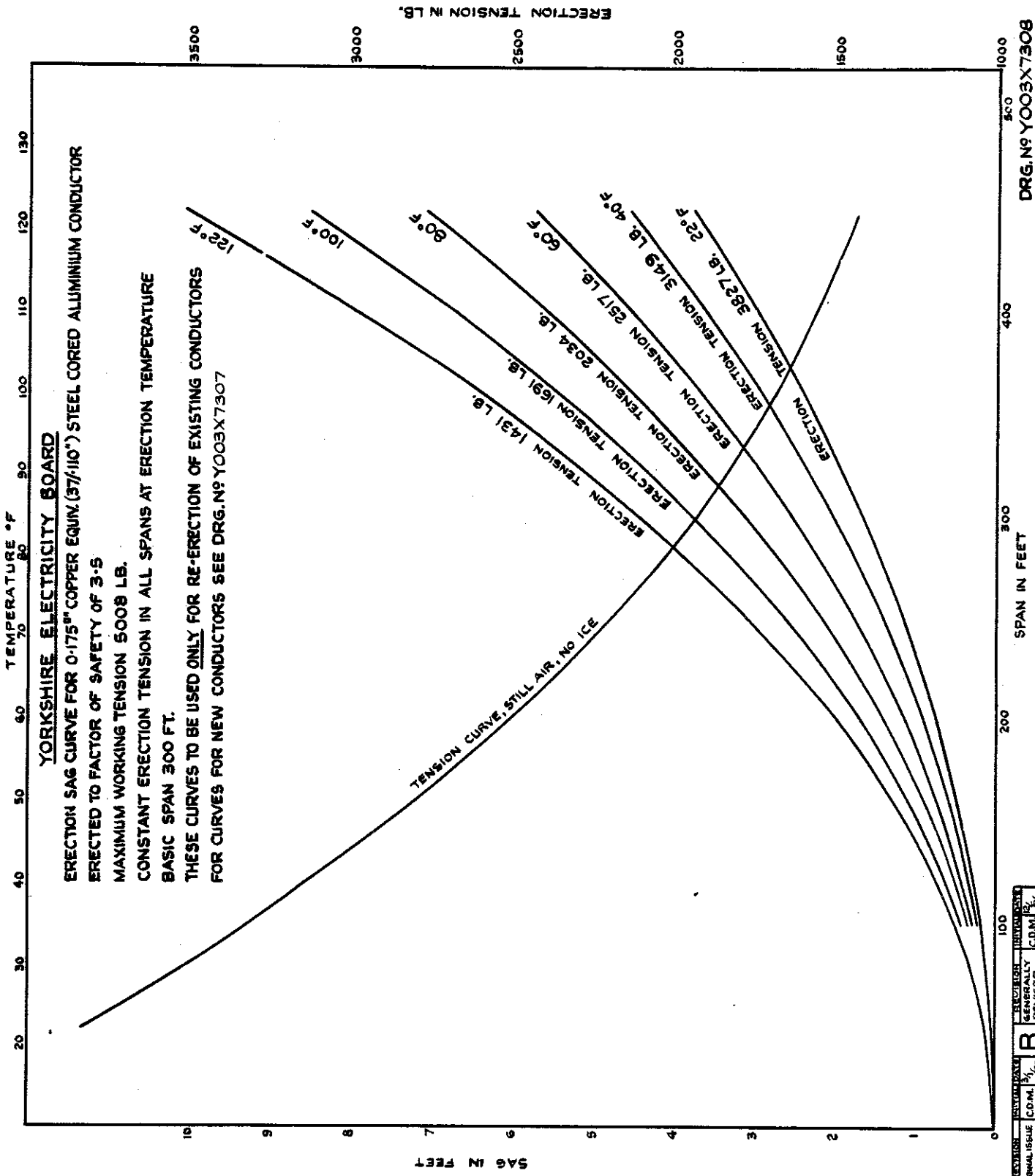
NOTE:- THESE CURVES TO BE USED ONLY FOR THE
ERECTION OF NEW CONDUCTORS.
FOR RE-ERECTION OF EXISTING CONDUCTORS
SEE DRG NO Y003X7304



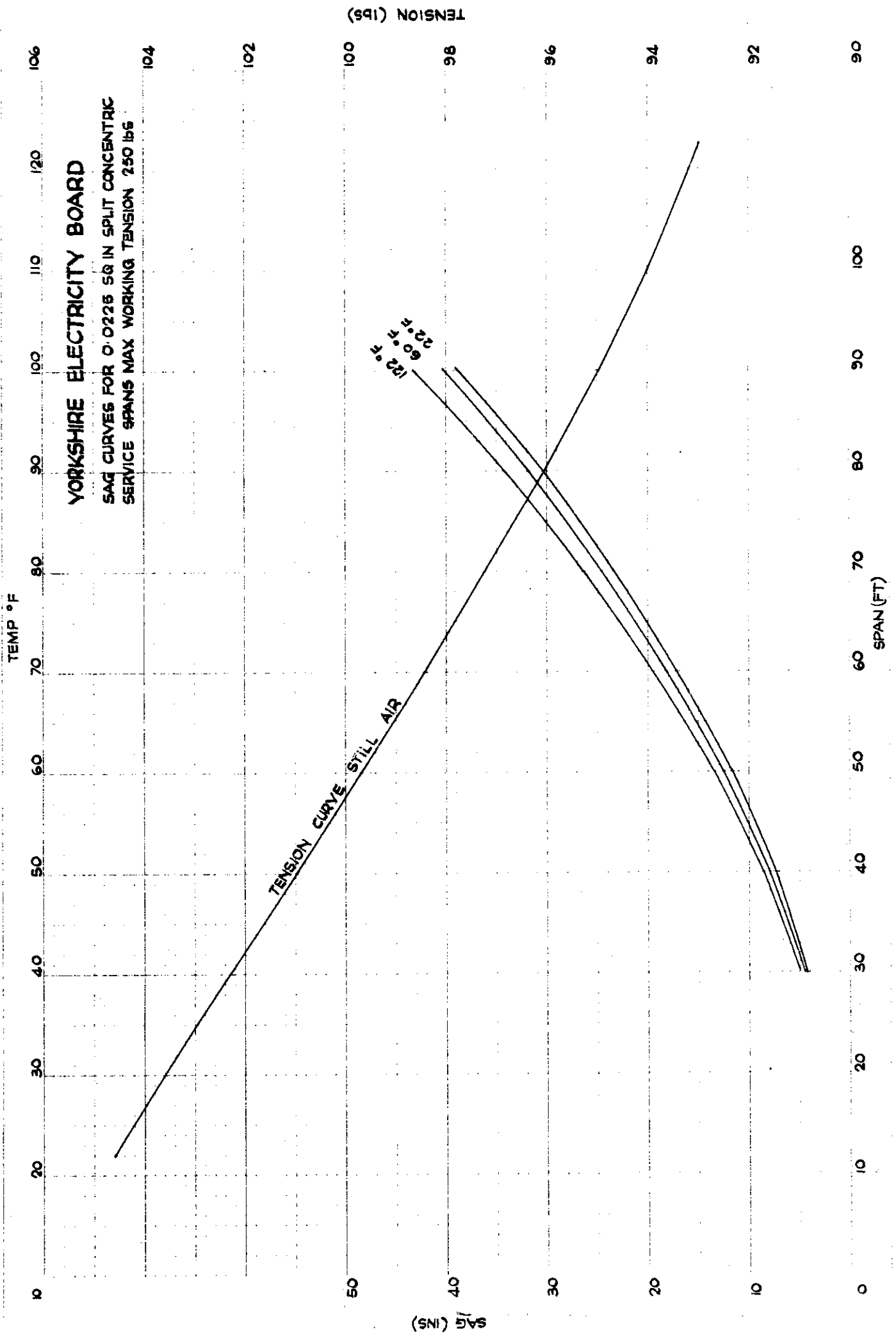


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